

| Question |     |       |    | Marking details                                                                                                                                      | Marks available |          |          |           |          |          |
|----------|-----|-------|----|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |    |                                                                                                                                                      | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 7        | (a) |       |    | pH                                                                                                                                                   |                 |          | 1        | 1         |          | 1        |
|          | (b) | (i)   |    | Scale correct with 0 or 10 at origin. Must cover more than half the grid                                                                             | 1               |          |          | 1         | 1        |          |
|          |     | (ii)  |    | All plots correct , ( +/- 1 small square)<br>(one error = 1 mark two/more errors =0)                                                                 |                 | 2        |          | 2         | 2        |          |
|          |     | (ii)  |    | Quality of line                                                                                                                                      |                 | 1        |          | 1         | 1        |          |
|          | (c) | (i)   |    | Increase (1)<br>More (successful) collisions /OWTTE (1)                                                                                              |                 | 1        | 1        | 2         |          |          |
|          |     | (ii)  |    | Denaturation/ altering shape of {active site/ enzyme}/ enzyme destroyed                                                                              | 1               |          |          | 1         |          |          |
|          |     | (iii) | I  | 40/Consistent with graph as drawn                                                                                                                    |                 | 1        |          | 1         | 1        |          |
|          |     |       | II | Insufficient temperatures/no intermediate values (1)<br>Increase number of temperatures tested {in 35°– 45° C range/<br>near optimum/ around 40} (1) |                 |          | 2        | 2         |          | 2        |
|          |     |       |    | <b>Question 7 total</b>                                                                                                                              | <b>2</b>        | <b>5</b> | <b>4</b> | <b>11</b> | <b>5</b> | <b>3</b> |